

Uncertainties in the 20th century carbon budget associated with land use change

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Abstract

Uncertainties in the 20th century carbon budget associated with the treatment of land use change (LUC) are assessed using the Canadian Centre for Climate Modelling and Analysis (CCCma) first-generation Earth System Model (CanESM1). Eight coupled climate carbon cycle simulations are performed using different reconstructions of 1850–2000 land cover derived from historical information on changes in cropland and pasture area. The simulations provide estimates of the emissions associated with LUC, the relative contribution of changes in cropland and pasture to LUC emissions and the uncertainty associated with differences among historical data sets of crop area as well as in the manner in which the historical land cover data are constructed. The resulting estimates of the amount of biomass deforested over the 1850–2000 period range from 63 to 145 Pg C with cumulative implied LUC emissions ranging from 40 to 77 Pg C. These values of LUC emissions are considerably lower than Houghton's estimate of 156 Pg C. The year 2000 atmospheric CO₂ concentration ranges between 371.1 ± 3.7 ppm depending on the data set used and the manner in which historical land cover is constructed. This compares to the observed value of 369.6 ppm at Mauna Loa and is 17.3 ± 6.3 ppm larger than for simulations without LUC. Although increases in cropland result in the expected increase in LUC emissions, changes in pasture area decrease these emissions because of carbon sequestration in soils.

Keywords: carbon budget, earth system models, land use change, terrestrial ecosystem models

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Introduction

Anthropogenic land use change (LUC) has been an important contributor to the historical carbon budget. For the 1850–2000 period, emissions of CO₂ due to LUC have contributed as much as 156 Pg C to the atmospheric CO₂ budget according to Houghton & Hackler (2001) compared with around 300 Pg C from fossil fuel burning. Historically, LUC has been driven by the need for crop and pasture area to meet the cereal and meat demands of an increasing population. The fraction of land area under crops and pastures has increased from roughly 7% in 1850 to about 33% in 1990 (Ramankutty & Foley, 1999; Klein Goldewijk, 2001).

LUC affects the regional and global climate through both biophysical and biogeochemical pathways. For example, the biophysical characteristics of the land surface including albedo, roughness length, rooting depth and leaf area index (LAI) are all modified when natural vegetation is replaced by croplands. These changes alter the partitioning of net radiation between latent and sensible heat fluxes and the partitioning of

precipitation between runoff and evapotranspiration. The associated feedbacks result in changes to regional and global climate (Bonan *et al.*, 1992; Betts, 2000; Chase *et al.*, 2000; Douville *et al.*, 2000; Heck *et al.*, 2001; Negri *et al.*, 2004). It is estimated that the biophysical effects of anthropogenic LUC would, if acting alone on the climate system, cause a net cooling of around 0.12 °C (Brovkin *et al.*, 2004; Matthews *et al.*, 2004) equivalent to radiative forcing of about -0.15 to -0.20 W m⁻² (Hansen *et al.*, 1998; Betts, 2001; Matthews *et al.*, 2003) primarily because of an increase in albedo when natural vegetation is replaced by croplands.

The associated biogeochemical aspects of LUC are not well quantified. Deforestation (the term is used here to indicate the removal of all sorts of natural vegetation including forests and grasslands) leads to the release of CO₂ through the destruction of standing biomass and its subsequent decomposition. When land is deforested, part of the carbon in the standing biomass is immediately released to the atmosphere by burning while the remainder is released at longer time scales through the decay of waste and, ultimately, of products manufactured from the deforested biomass. Land that is converted to agricultural uses stores much less carbon in its standing biomass than does the forest it replaces.

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Storage as soil carbon is also reduced because agricultural tillage of croplands enhances soil carbon decomposition through increased aeration. Deforested areas generally have lower amounts of biomass in their vegetation and soil carbon pools (Fearnside, 2000). The release of CO₂ associated with LUC contributes to climate warming but the magnitude of the warming depends on the fraction of the emissions that remains in the atmosphere. This in turn depends on CO₂ uptake by terrestrial and oceanic physical and biological processes.

Biogeochemical aspects of LUC are usually investigated in coupled climate carbon cycle models by injecting the CO₂ emissions associated with LUC directly into the atmosphere in the same manner as for fossil fuel emissions. This is done in C⁴MIP simulations, for instance (Friedlingstein *et al.*, 2006). The direct injection of LUC emissions into the atmosphere of coupled climate carbon cycle models, with no corresponding changes at the land surface, does not take into account the associated albedo changes or the reduced capacity of the biosphere to sequester carbon, especially when natural vegetation is replaced with croplands. Matthews *et al.* (2004) and Brovkin *et al.* (2004) specify LUC emissions from Houghton (2003) but also change the land cover at the surface in order to incorporate the biophysical effects of an increase in crop area in a partially consistent manner. Matthews *et al.* (2004) estimate that LUC from 1850 to 2000 has resulted in an additional 26 ppmv of CO₂ in the atmosphere and that this increase in CO₂ has caused a warming of 0.15 °C, which is slightly larger than their estimate of a 0.13 °C biophysical cooling due to the associated increase in albedo. Brovkin *et al.* (2004) estimate an increase in atmospheric CO₂ of between 22 and 43 ppmv due to LUC emissions from 1800 to 2000 depending on the terrestrial ecosystem model (TEM) and crop data set used.

The LUC-related CO₂ emissions used in coupled climate carbon cycle models are currently determined using two approaches. The first assigns the amount of biomass deforested to several pools which return carbon to the atmosphere based on different specified decomposition rates, following a 'book-keeping approach' such as that of Houghton (2003). This approach tracks LUC emissions due to changes in cropland and pasture area, shifting cultivation, wood harvesting and logging but does not take into account the effect of CO₂ or climate change on vegetation growth or that deforested patches have less potential to sequester carbon as CO₂ increases with time.

These limitations are partially addressed in the second approach that estimates LUC emissions in the framework of TEMs. The TEMs are driven offline with specified atmospheric CO₂ concentration and climate

data and the difference between net atmosphere to land CO₂ flux for cases with and without LUC gives the inferred LUC emissions. McGuire *et al.* (2001) report LUC emissions estimated by four TEMs and Jain & Yang (2005) that from one TEM. In these studies, TEMs are driven with specified increasing CO₂ and varying climate. Both studies use the Ramankutty & Foley (1999) (hereafter referred to as RF99) crop data set for estimating deforestation. Jain & Yang (2005) use Houghton's regional estimates of deforested area in addition to RF99. The difficulty here is that forcing the no-LUC simulations with observed values of CO₂, which necessarily include the effects of LUC, is a form of double counting which yields higher carbon uptake by land than would have occurred in the absence of LUC. The result is an overestimate of LUC emissions when differencing simulations with and without LUC. In a related offline approach, Shevliakova *et al.* (2009) investigate uncertainties in LUC emissions using four land cover data reconstructions from Hurtt *et al.* (2006) in a TEM driven with constant specified CO₂ of 350 ppm. The land cover reconstructions are based on RF99 crop data, the HYDE pasture and crop data (Klein Goldewijk, 2001) and assumptions about wood-harvesting practices and patterns. Shevliakova *et al.* (2009) use the 350 ppm CO₂ concentration to initialize their pre-industrial (year 1700 in their case) vegetation and soil carbon pools as well as to perform transient simulations for the 1700–2000 period. Here the use of CO₂ concentration of 350 ppm for initializing preindustrial carbon pools overestimates their magnitude so that LUC emissions will be overestimated when these pools are subjected to deforestation and wood harvesting.

The approach considered here uses anthropogenic CO₂ emissions and changing land cover for the 1850–2000 period to drive a fully coupled climate carbon cycle model in which CO₂ is a prognostic variable. The LUC emissions over the 1850–2000 period are due to the replacement of natural vegetation by croplands and pastures. The LUC emissions are obtained interactively and surface albedo changes as well as changes in terrestrial and oceanic CO₂ fluxes and storage all interact with and affect one another. Global data sets of the historical increase in cropland area are from RF99 and crop and pasture data from Klein Goldewijk (2001) (the HYDE 2 data set) are also used. Simulations are performed for increases in croplands based on the two data sets, for an increase in pasture area alone and for increases in both pasture and croplands together in order to identify the effects of changes in crop and pasture area separately. LUC is implemented using two approaches for each case resulting in eight 1850–2000 land cover reconstructions. The consequences for simulated atmospheric CO₂ and for land–atmosphere

and ocean–atmosphere CO₂ exchange for all eight land cover reconstructions are investigated using a comprehensive earth system model (ESM). To our knowledge, this represents the first attempt to investigate uncertainty in LUC emissions and the effect on atmospheric and land carbon budget within such a framework.

The carbon budget and LUC

CanESM1

CanESM1 is a comprehensive coupled carbon–climate model based on Canadian Centre for Climate Modelling and Analysis (CCCma's) third-generation atmospheric general circulation model (AGCM3) and described in detail by Arora *et al.* (2009). The physical ocean component (OGCM3.5) of CanESM1 is based on the National Center for Atmospheric Research (NCAR) community ocean model (NCOM1.3) (Gent *et al.*, 1998) which is a primitive equation model with a rigid lid. There is no flux adjustment. The preindustrial carbon state of CanESM1 is analyzed by Christian *et al.* (2010) and an analysis of the simulated 20th century carbon budget is presented in Arora *et al.* (2009).

The vertically integrated carbon budget equations for the ocean, land and atmospheric components of the climate system as modelled here are

$$\frac{dH_A}{dt} = A - F_O - F_L + E, \quad (1a)$$

$$\frac{dH_L}{dt} = F_L = G - R, \quad (1b)$$

$$\frac{dH_O}{dt} = A_O + F_O, \quad (1c)$$

where $H = H_A + H_L + H_O$ represents the total carbon burden in the column and the carbon burden of the Atmosphere, Land and Ocean components as indicated by the subscripts. The total amount of carbon in the system increases due to anthropogenic fossil fuel emissions E into the atmosphere. All quantities are functions of geographical location and time. The emitted carbon is redistributed by transport processes A and A_O in the atmosphere and the ocean and by fluxes F_L , F_O between atmosphere and land and the atmosphere and ocean. The atmosphere–land flux $F_L = G - R$ is the difference between G the carbon gain via photosynthesis [the gross primary productivity (GPP)] and R the respiration. For an equilibrium 1850 control simulation $E = 0$ and the globally averaged climatological means of terms in Eqn (1) are near zero.

Atmospheric CO₂ is a fully prognostic three-dimensional tracer in CanESM1. There are no sources or sinks of CO₂ in the body of the atmosphere and carbon enters and leaves the atmosphere in the form of anthropogenic emissions E and fluxes to or from the underlying land and ocean. The biological and inorganic ocean carbon component of CanESM1 is the Canadian Model of Ocean Carbon (CMOC) which incorporates an inorganic chemistry module (solubility pump) and an ecosystem model (organic and carbonate pumps) (Zahariev *et al.*, 2008). Terrestrial ecosystem processes in CanESM1 are modelled using the Canadian TEM (CTEM) (Arora, 2003; Arora & Boer, 2003, 2005). CTEM simulates three live vegetation pools (leaves, stem and root) and two dead carbon pools (litter and soil organic carbon) for nine plant functional types (PFTs) as shown in Fig. 1. CTEM is coupled to the Canadian Land Surface Scheme (CLASS 2.7) (Verseghy, 1991; Verseghy *et al.*, 1993) and the two components together produce fluxes of energy, water and CO₂ at the land–atmosphere boundary in CanESM1. The photosynthesis submodule of CTEM is based on the biochemical model of Farquhar *et al.* (1980) and Collatz *et al.* (1991, 1992). The current version uses a single-leaf photosynthesis approach with coupling between photosynthesis and canopy conductance based on vapour pressure deficit (Leuning, 1995).

In CTEM, the budget equations for carbon in the vegetation (V), litter or debris (D) and soil carbon (S) pools in Fig. 1 are written as

$$\frac{dH_V}{dt} = G - R_V - I_{VD}, \quad (2a)$$

$$\frac{dH_D}{dt} = I_{VD} - R_D - I_{DS}, \quad (2b)$$

$$\frac{dH_S}{dt} = I_{DS} - R_S, \quad (2c)$$

and their sum yields Eqn (1b) which connects the land carbon budget to the overall carbon budget. Here $H_L = H_V + H_D + H_S$ is the total land carbon (kg C m⁻²) comprising of vegetation, litter and soil carbon pools. $R = R_V + R_D + R_S$ where R_V is the vegetation growth and maintenance respiration and R_D and R_S the heterotrophic respiration from the litter and the soil carbon pools. I_{VD} is the transfer of litter from the vegetation to the litter pool and I_{DS} the humidification of litter to soil carbon.

LUC emissions

The replacement of natural vegetation by crops and pastures affects all components in (1) interactively and

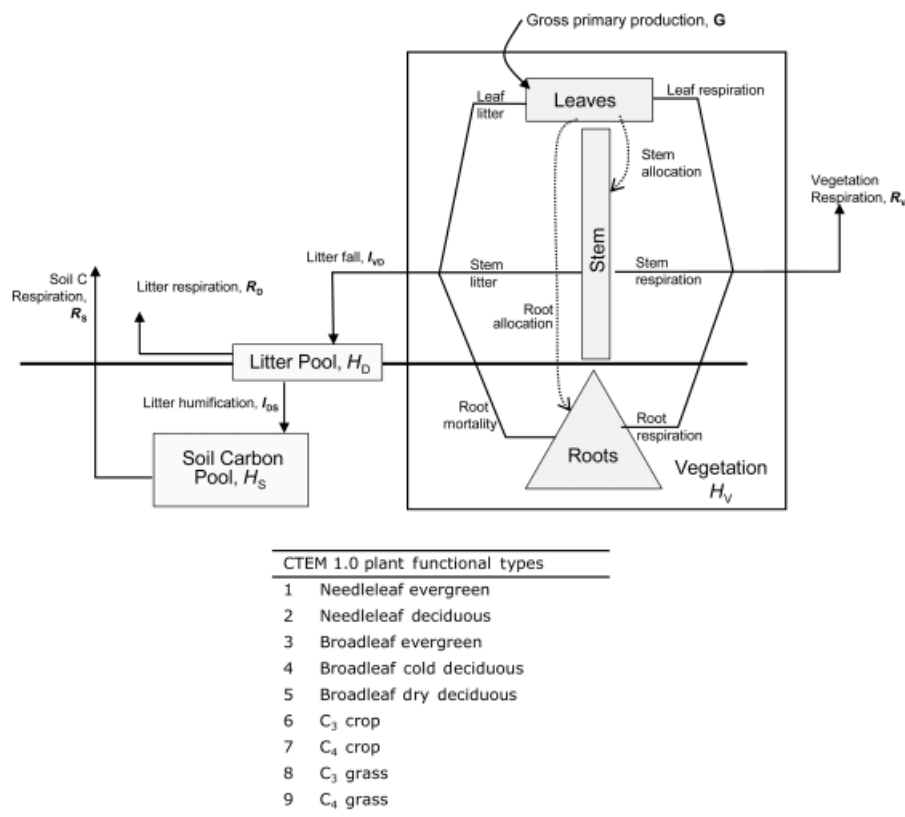


Fig. 1 Structure of the Canadian Terrestrial Ecosystem Model (CTEM) and the nine plant functional types for which all ecosystem processes are modelled.

has historically led to an increase in atmospheric CO₂ (or equivalently H_A). LUC effects are sometimes introduced as an 'additional emission' term E_L , estimated externally to the model and prescribed in Eqn (1a) in the same manner as the fossil fuel emissions E . This approach, however, compromises the interactivens and interconnectedness of all of the components in (1). Other approaches prescribe the atmospheric CO₂ concentration and LUC while allowing other components of the system to react (e.g. McGuire *et al.*, 2001) but this also compromises the balances among the carbon pools since the atmospheric pool is prescribed. In the approach adopted here, we introduce LUC in a fully interactive way and obtain the resulting LUC emissions as the difference between atmosphere to land CO₂ flux for simulations with and without LUC.

An increase in crop or pasture area implies deforestation and a change in H_V through the replacement of natural herbaceous and/or woody vegetation. The deforested biomass ($L = L_A + L_D + L_S$) is divided into three parts, the amounts combusted or used for fuel wood with no time delay which contributes to atmospheric CO₂ (L_A), the component left as slash or used for pulp and paper products (L_D) and that used for durable

wood products (L_S). The fractions allocated to these three uses depend on the aboveground vegetation biomass density and whether the PFTs are woody or herbaceous (Table 1). To account for the timescales involved, the fraction allocated to slash or pulp and paper products (L_D) is transferred to the litter pool and the fraction allocated to wood products (L_S) to the model's soil carbon pool. LUC redistributes carbon from live vegetation to dead litter and soil carbon pools and emits CO₂ to the atmosphere through direct burning of the deforested biomass (L_A). The result is a perturbation to all the terms in the budgets and a modification of the flux and storage terms over land because of the changes in vegetation, litter and soil carbon pools. The corresponding land carbon budget equation with LUC becomes

$$\frac{dH_L^*}{dt} = F_L^* = G^* - R^* - L_A. \quad (3)$$

The difference between atmosphere to land CO₂ flux F_L between the reference no-LUC case and the flux F_L^* in the LUC case gives the LUC emissions to the atmosphere (E_L) as the net decrease in atmosphere to land

Table 1 Fractions of aboveground deforested biomass that are (1) combusted or used for fuel wood, (2) turned into slash or used for pulp and paper products and (3) used for durable wood products, when fractional coverage of vegetation decreases through LUC

	Fraction of deforested biomass that is ...		
	combusted or used for fuel wood (ϵ_A)	turned into slash or used for pulp and paper products (ϵ_D)	used for long life wood products (ϵ_S)
<i>Woody PFTs</i>			
Aboveground biomass (kg C m^{-2}) > 4.0	0.15	0.7	0.15
1.0 < aboveground biomass (kg C m^{-2}) < 4.0	0.30	0.7	0.00
Aboveground biomass (kg C m^{-2}) < 1.0	0.45	0.55	0.00
<i>Herbaceous PFTs</i>	0.45	0.55	0.00

LUC, land use change; PFTs, plant functional types.

CO₂ flux

$$\begin{aligned} E_L &= F_L - F_L^* \\ &= L_A + (R^* - R) - (G^* - G), \end{aligned} \quad (4)$$

with all components of the coupled carbon model active in both simulations. When a region is deforested, L_A is positive and, as well, respiration R^* is greater than R due to the decomposition of deforested biomass. The change in gross photosynthetic uptake ($G^* - G$) is comparatively small. When croplands are abandoned and reforestation is occurring, L_A is zero, and regrowth of forests means that $G^* - R^*$ is greater than $(G - R)$ and the forests are sequestering carbon so that E_L can become negative in this case. In regions where pasture lands expand E_L also becomes negative because of lower R^* associated with low soil decomposition rates for grasslands.

The maximum turnover rates for litter ($\sim 0.5 \text{ year}^{-1}$) and soil carbon ($\sim 0.025 \text{ year}^{-1}$) are modified by temperature and moisture in CTEM and depend interactively on the simulated climate. Houghton's book-keeping approach (Houghton & Hackler, 2001), by comparison, divides the deforested biomass into slash (with turnover time of less than a year) and pools with turnover rates of 1 year^{-1} (fuel wood), 0.1 year^{-1} (pulp and paper products), 0.01 year^{-1} (wood products) and 0.001 year^{-1} (elemental carbon) that are not climate dependent.

A simple crop model is used over the cropland fraction of a grid cell. Crops increase their biomass depending on environmental conditions with harvesting initiated when the air temperature remains below 8°C for 5 consecutive days, or when the crop LAI reaches a threshold (3.5 for C_3 crops and 4.5 for C_4 crops) signifying that the crops have matured (Arora & Boer, 2005). Crops are harvested over a period of 15 days and the harvested biomass contributes to the litter

pool. This typically leads to one annual crop cycle in high- to mid-latitude regions and multiple crop cycles in tropical regions. Harvesting ensures that vegetation biomass does not keep increasing on croplands as CO₂ increases, and prevents croplands from sequestering carbon like forests. Carbon may still be sequestered in the soil carbon pool but the higher soil carbon decomposition rate for croplands ($\sim 0.03 \text{ year}^{-1}$) compared with other PFTs ($0.013\text{--}0.026 \text{ year}^{-1}$), which is due to tillage, prevents soil carbon sequestration in croplands. CTEM does not track the age of vegetation explicitly but characterizes the state of the vegetation in terms of the mean values of carbon in the various pools. When croplands or pasture areas are abandoned, the fractional coverage of other PFTs is increased according to the land cover reconstructions. The result is that the vegetation biomass for a PFT is spread over a larger area reducing vegetation density. Carbon is sequestered until a new equilibrium is reached providing a carbon sink associated with regrowth as the abandoned areas revert back to natural vegetation. The C_3 and C_4 grasses in the model are characterized by lower soil carbon decomposition rates ($\sim 0.013 \text{ year}^{-1}$) compared with other PFTs ($0.02\text{--}0.03 \text{ year}^{-1}$) which leads to higher soil carbon densities over grasslands as is observed in nature and highlighted in studies illustrating the carbon sequestering potential of grasslands (Schlesinger, 1977; Guo & Gifford, 2002; Jackson *et al.*, 2002).

In summary, the burning and subsequent decomposition of the biomass lost when cropland and pasture area increase, the growth and harvesting of crops, the regrowth of vegetation on abandoned crop and pasture areas, and the differences in soil carbon decomposition rates for croplands and grasslands compared with the woody PFTs they replace all contribute to LUC-related carbon emissions and affect atmospheric CO₂ in fully interactive simulations.

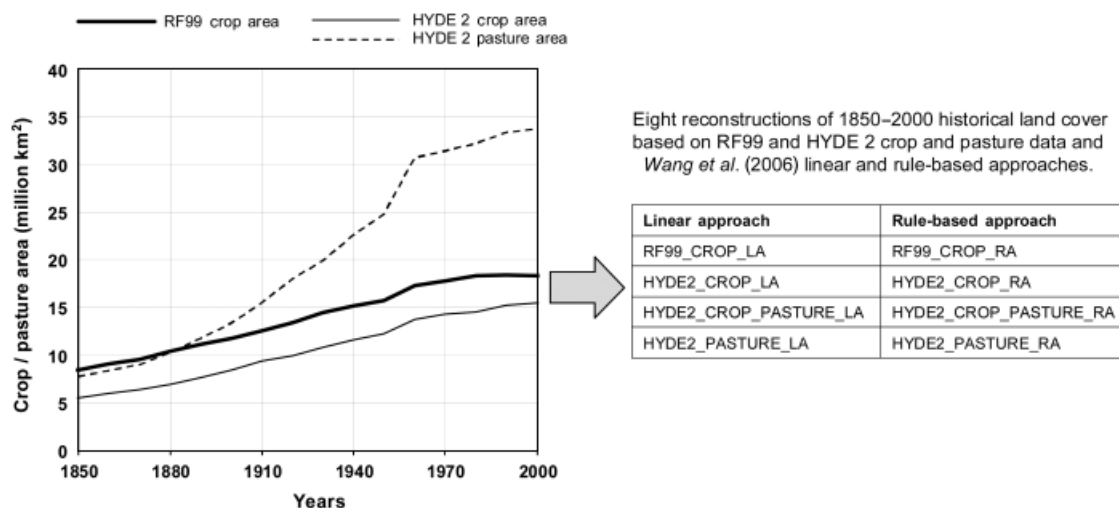


Fig. 2 Historical crop area in Ramankutty & Foley (1999) (RF99) and crop and pasture areas in HYDE 2 data sets (left panel). These data sets are used to obtain eight reconstructions of historical land cover with increases in crop area only, pasture area only and both crop and pasture areas.

Land cover data sets

Two different crop and pasture data sets guide the reconstruction of historical land cover maps that provide the fractional coverage of the nine PFTs used in CTEM. The RF99 data set provides the fractional coverage of cropland at 0.5° resolution for the 1700–1992 period. The HYDE 2 data set provides cropland and pasture information at 30'' resolution for the 1700–2000 period (Klein Goldewijk, 2001).

Wang *et al.* (2006) generate fractional coverage of CTEMs nine PFTs from 1850 to 1992 based on RF99 crop area and from 1993 to 2000 based on crop area results from the IMAGE 2 model (Alcamo *et al.*, 1998). The 1850 land cover in Wang *et al.* (2006) is derived from the IGBP DISCover data set with the regions dominated by present-day human land use replaced by modelled potential natural vegetation classes taken from the BIOME3 data set of Haxeltine & Prentice (1996). From 1850 onwards, Wang *et al.* (2006) use linear and rule-based approaches to generate land cover data sets as discussed in some detail in Appendix S1. In brief, in the rule-based approach an increase in crop area within a grid cell is first met by reducing the area covered by herbaceous PFTs (C_3 and C_4 grasses). If the increase in crop area cannot be met from the area covered by herbaceous PFTs it is obtained by removal of woody PFTs. The linear approach reduces the fractional coverage of herbaceous and woody PFTs in proportion to their existing fractional coverage in order to allow an increase in crop fraction. Simultaneous deforestation of both herbaceous and woody PFTs in the linear approach leads to more deforested biomass compared with the

rule-based approach where the herbaceous PFTs are deforested first. These approaches are also applied to the HYDE 2 crop and pasture data. Pastures are treated as a subset of grasslands.

The global crop area obtained using the HYDE 2 data set is smaller than that based on the RF99 data as seen in Fig. 2 which also shows a large increase in pasture area in the HYDE 2 data. The RF99 and HYDE 2 data sets are used to obtain eight reconstructions of 1850–2000 land cover for increases in crop area, pasture area and both together using the linear (LA) as well as the rule-based (RA) approaches (Fig. 2). These approaches are applied at 0.5° spatial resolution to obtain land cover reconstructions (in terms of fractional coverages of CTEM PFTs) and then aggregated to the 3.75° resolution of CanESM1. Figure 3 shows the change from 1850 to 2000 in crop (ΔC), grass (ΔH) and woody (ΔW) fractions at the 3.75° resolution of CanESM1 for the RA case. Their sum $\Delta C + \Delta H + \Delta W = \Delta B$ is generally close to 0, where ΔB is the change in the bare ground fraction that is not covered with vegetation. For cases with increases in pasture area, however, some demand for pasture area has to be met by decreasing the bare ground fraction. In the left-hand column of Fig. 3, the increase in crop fraction over North America, Eastern Europe, India, South-Eastern China and southern South America is apparent in reconstructions based on both RF99 and HYDE 2 data sets (first two rows). Note also the decrease in crop area in a few grid cells in the Eastern United States in both data sets due to the abandonment of croplands which results in an increase in the woody fraction by the year 2000. Figure 3b and e show the corresponding changes in grass fraction and Fig. 3c and

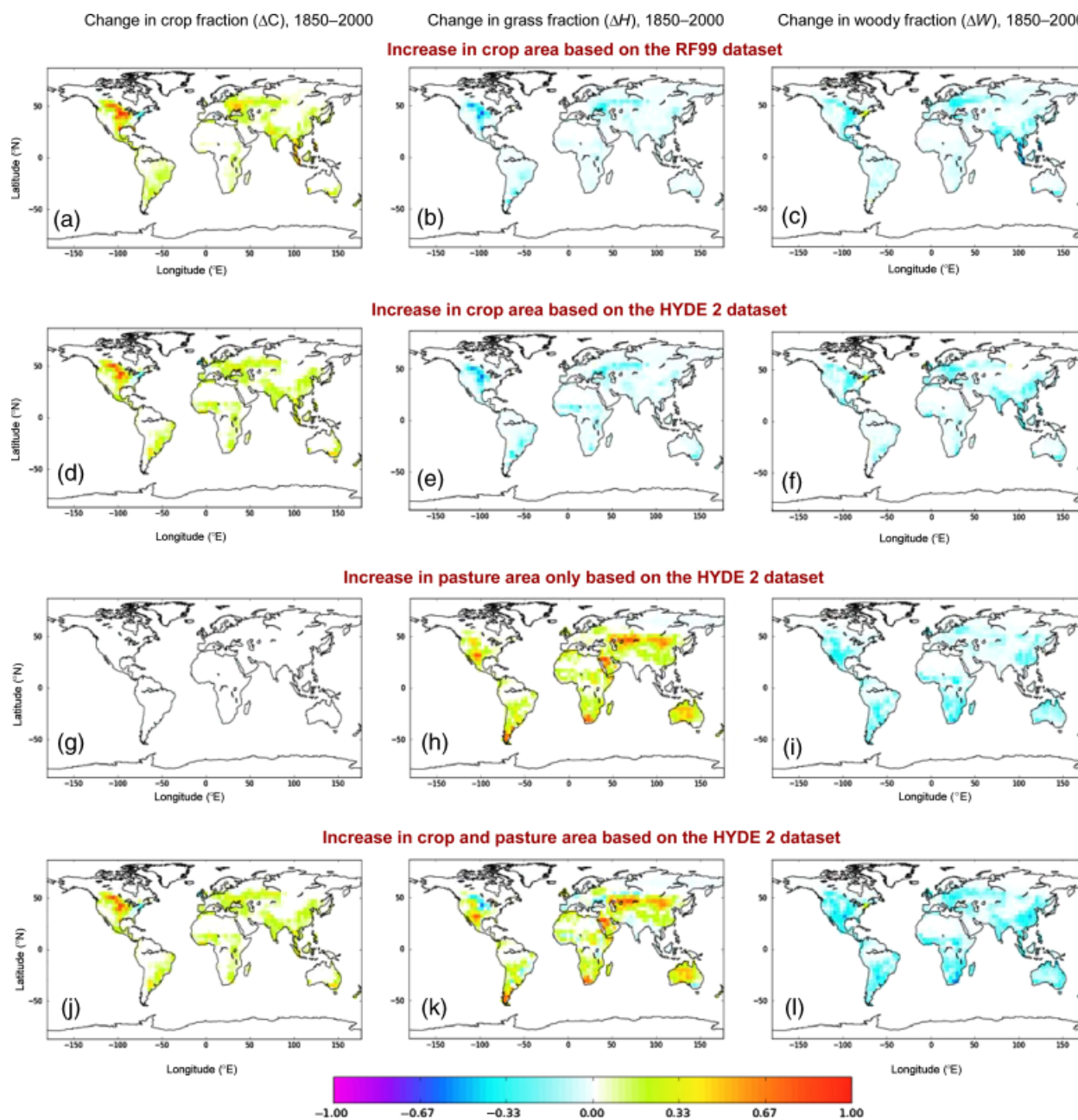


Fig. 3 Fractional changes in coverage of crop (left column), grass (middle column) and woody (right column) plant functional types (PFTs) from 1850 to 2000, for increases in crop area (RF99 and HYDE 2), pasture area (HYDE 2) and both crop and pasture area (HYDE 2) scenarios, obtained using the rule-based approach.

show the changes in woody fraction. These are mostly reductions because of the increase in crop area. Figure 3h shows the increase in fractional coverage of grasses associated with increases in pasture area only following the HYDE 2 reconstruction. There are no changes in crop area in this case (Fig. 3g) but the woody (Fig. 3i) and bare (not shown) fractions decrease to incorporate increases in pasture area. Finally, the reconstruction

based on the HYDE 2 crop and pasture data sets yields changes in the fractional coverage of both crops and grasses (Fig. 3j and k). The corresponding reductions in the woody fraction (Fig. 3l) are largest in this case. This reconstruction yields the largest amount of deforested biomass, but also a partially compensating sequestration of soil carbon in grasslands as discussed in 'Global values'.

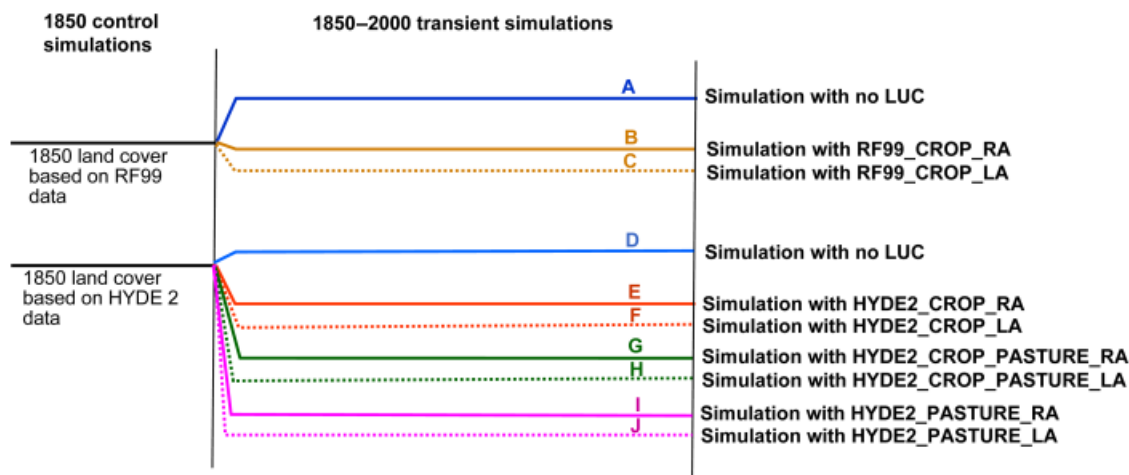


Fig. 4 Schematic showing the 10 transient 1850–2000 simulations performed: two with no land use change and eight with different reconstructions of the historical land cover based on RF99 and HYDE 2 data sets. The transient simulations are initialized from 1850 control simulations.

The characterization of LUC as an increase in grassland area in Fig. 3 depends to some extent on the manner in which land cover is reconstructed. Here, we have assumed that pastures are grasslands and that a historical increase in pasture area implies an increase in the area of grasslands. However, an increase in pasture area may not always imply a change in land cover but may instead reflect a change in land use. For instance, an increase in grazing activity in a region where the primary vegetation is shrubs may be classified as an increase in pasture area but need not imply a transformation of shrubs to grassland. The treatment of pastures as grasslands in our study represents an upper limit on grassland increase due to LUC. A land cover reconstruction that takes into account that a historical increase in pasture area can represent changes in both land cover and land use will result in lower increase in grassland area and therefore less carbon sequestration in soil carbon.

The motivation behind using land cover reconstructions with increases in crop and pasture area separately, and crop and pasture areas together, is to diagnose the contribution of pastures to LUC emissions. The lack of an increase in pasture area in the McGuire *et al.* (2001) study has been suggested by House *et al.*, (2003) to be the reason for their lower LUC emissions compared with those of Houghton (2003). However, Jain & Yang (2005), who perform simulations with both RF99 based crop data and Houghton's regional deforestation estimates, show that Houghton's estimates are higher mainly because of the difference in the rate of increase of cropland area and not the increase in pasture area. Strassmann *et al.* (2008) suggest that increases in pasture

area based on the HYDE data set more than doubles the cumulative LUC emissions from 84 to 188 Pg C for the 1700–1999 period. Simulations performed by Shevliakova *et al.* (2009) give a contribution from pastures ranging from a sink of 0.15 to a source of 0.37 Pg C yr⁻¹ for the 1990s.

Experimental approach

Ten fully coupled climate-carbon cycle simulations are performed for the 1850–2000 period all forced with the same anthropogenic fossil fuel and industrial CO₂ emissions. Two of the simulations do not include LUC and the others use one of eight land cover reconstructions (Fig. 4). The differences between the net atmosphere to land CO₂ flux in these simulations yields the LUC emissions [Eqn (4)]. The simulations without LUC use 1850 land cover throughout the 1850–2000 period. The 1850 land cover is slightly different depending on whether it is reconstructed using 1850 crop data from RF99 or the crop and pasture data from the HYDE 2 data set. The primary difference in 1850 between the two land cover reconstructions is a larger crop area in the reconstruction based on RF99. These 1850 land cover reconstructions are also used in preindustrial control simulations in which CO₂ varies around 288 ppm and anthropogenic emissions are absent. The 10 transient simulations in Fig. 4 are initialized from their corresponding preindustrial control simulations.

CO₂ is a three-dimensional tracer in these transient simulations and its concentration depends on spatially distributed anthropogenic CO₂ emissions, atmospheric transport and the interactive land- and

ocean–atmosphere exchange of CO₂. The anthropogenic CO₂ emissions used are based on the EDGAR V1.3 (1890–1960) and EDGAR V3.2 (1970–1990 and 1995) data sets (van Aardenne *et al.*, 2001) (<http://www.mnp.nl/edgar/model/>) as explained in Arora *et al.* (2009). The historical concentrations of non-CO₂ GHGs including CH₄, N₂O, CFC-11 and CFC-12 are prescribed as global means.

Results

Preindustrial terrestrial biosphere state

Table 2 compares the net primary productivity (NPP) and GPP and the vegetation, litter and soil carbon pools simulated during the 1850s with other model-based estimates. Table 2 also compares observation and model-based estimates of these quantities, which represent the late 20th century values, with simulated results for the 1981–2000 period. CanESM1 results are given for simulations using both RF99 and HYDE 2 land cover reconstructions. The vegetation and soil carbon masses are slightly lower during the 1850s for the simulation that uses land cover based on the RF99 data set compared with that using the HYDE 2 based land cover reconstruction because of the larger crop area in the RF99 data set. Croplands contain less vegetation biomass because they are frequently harvested and less soil carbon values because of the higher decomposition rates associated with tillage. Simulated NPP and GPPs and global vegetation biomass and litter compare well with observation-based estimates in Table 2, although the simulated soil carbon mass is somewhat high. The simulated NPP of 57 Pg C yr⁻¹ during 1850s can also be compared with values of between 50 and 70 Pg C yr⁻¹ from the C⁴MIP models (Friedlingstein *et al.*, 2006) integrated with preindustrial CO₂ of ~280 ppm. Figure 5 shows the preindustrial distribution of NPP and vegetation biomass for the simulation using RF99 based 1850 land cover. Modelled NPP and vegetation biomass exhibit the expected spatial structure with higher values in the warm and humid tropics and lower values at higher latitudes and in arid and semiarid regions. Modelled NPP values are somewhat lower in the Amazon region because of lower than observed precipitation (Arora *et al.*, 2009).

Global values

Deforested biomass. Figure 6a shows time series of deforested biomass (L) for the 1850–2000 period and Fig. 6b gives cumulative values ($\int L dt$). For simplicity, the results are averaged for scenarios using the linear and the rule-based approaches. The amount of

deforested biomass (L) does not equal the net LUC emissions E_L [Eqn (4)] as only a fraction L_A is transferred instantaneously to the atmosphere. E_L also includes contributions from other processes including crop growth and harvesting, regrowth of vegetation on abandoned crop and pasture areas, differences in soil carbon decomposition rates for croplands and grasslands, and differences in respiration from the carbon pools. Deforested biomass does, however, provide a straightforward measure of the difference between the several land cover reconstructions. The coloured bars in Fig. 6b show the range in cumulative deforested biomass by year 2000 that results from the use of linear and rule-based approaches. The cumulative values of deforested biomass vary from 63 to 146 Pg C by year 2000. The deforested biomass values for land cover reconstructions that are produced with the linear approach are higher than those produced with the rule-based approach. In the linear approach, an increase in crop or pasture area is satisfied by decreasing the area of other PFTs in proportion to their fractional coverage, whereas in the rule-based approach the less carbon-rich herbaceous PFTs are deforested first. The amount of deforested biomass is highest for simulations that include changes in both cropland and pasture area, as expected. The rate at which biomass is deforested decreases sharply after about 1960 in all cases (Fig. 6a) because crop and pasture area increase less rapidly in the RF99- and HYDE-based land cover reconstructions (e.g. Shevliakova *et al.*, 2009).

Land carbon stores. Figure 7 displays the simulated changes in total land carbon and in vegetation, litter and soil carbon components. Results are averaged for the cases using the linear and the rule-based approaches. For the simulations without LUC, the results are the average from simulations that use RF99- and HYDE-based 1850 land cover. Vegetation, litter and soil carbon all increase in the simulations without LUC because of the CO₂ fertilization effect. Arora *et al.* (2009) assess the CO₂ fertilization effect of CTEM in the framework of CanESM1 against observation-based and model values. The average increase in terrestrial carbon storage over the 1850–2000 period in simulations without LUC is 91 Pg C (Fig. 7a). The bars show the range of simulated changes in terrestrial storage by year 2000 when using RF99 and HYDE based 1850 land cover for the no-LUC simulations and when using linear and rule-based approaches for the simulations including LUC. When LUC is implemented, the terrestrial carbon storage initially decreases before recovering later in the century. In Fig. 7b the effect of LUC is to decrease

Table 2 Comparison of simulated terrestrial NPP, terrestrial GPP, vegetation biomass and litter and soil carbon pools during the 1850s and for the 1981–2000 period with observation-based and other model estimates

	Simulated 1850s		Simulated 1981–2000		Late 20th century values from modeling studies	
	Using land cover based on		Using land cover based on			
	RF99	HYDE	Preindustrial values from modeling studies	RF99		HYDE 2
Net primary productivity (NPP), Pg C yr ⁻¹	56.9	57.0	64.0 (Sitch <i>et al.</i> , 2003) 50–70 (Friedlingstein <i>et al.</i> , 2006)	67.5	69.4	60.0 (Ajtay <i>et al.</i> , 1979) 56.6 (Running <i>et al.</i> , 2004) 44.4–66.3 (Cramer <i>et al.</i> , 1999) 65.0 (Bonan <i>et al.</i> , 1992) (1980s and 1990s)
Gross primary productivity (GPP), Pg C yr ⁻¹	117.6	120.0	134.0 (Gerber <i>et al.</i> , 2004)	132.0	143.7	~125.0 (Zhao <i>et al.</i> , 2006) (MODIS derived LAI driven with NCEP reanalysis) 113.0 (Delire <i>et al.</i> , 2003)
Vegetation biomass, Pg C	544	563	923 (Sitch <i>et al.</i> , 2003)	541	541	560 (Ajtay <i>et al.</i> , 1979) 560 (Olson, 1983) 634 (Delire <i>et al.</i> , 2003)
Litter mass, Pg C	111	110	171 (Sitch <i>et al.</i> , 2003)	123	124	90 (Ajtay <i>et al.</i> , 1979)
Soil carbon mass, Pg C	1556	1613	1670 (Sitch <i>et al.</i> , 2003)	1558	1638	1395 (Post <i>et al.</i> , 1982) 1335 (Delire <i>et al.</i> , 2003) 1400–1600 (Schlesinger, 1977)

Results are shown for simulations using land cover reconstruction based on RF99 and HYDE 2 data sets. 1981–2000 results are averages of simulations that include increase in crop area (RF99) and both crop and/or pasture (HYDE 2) areas.

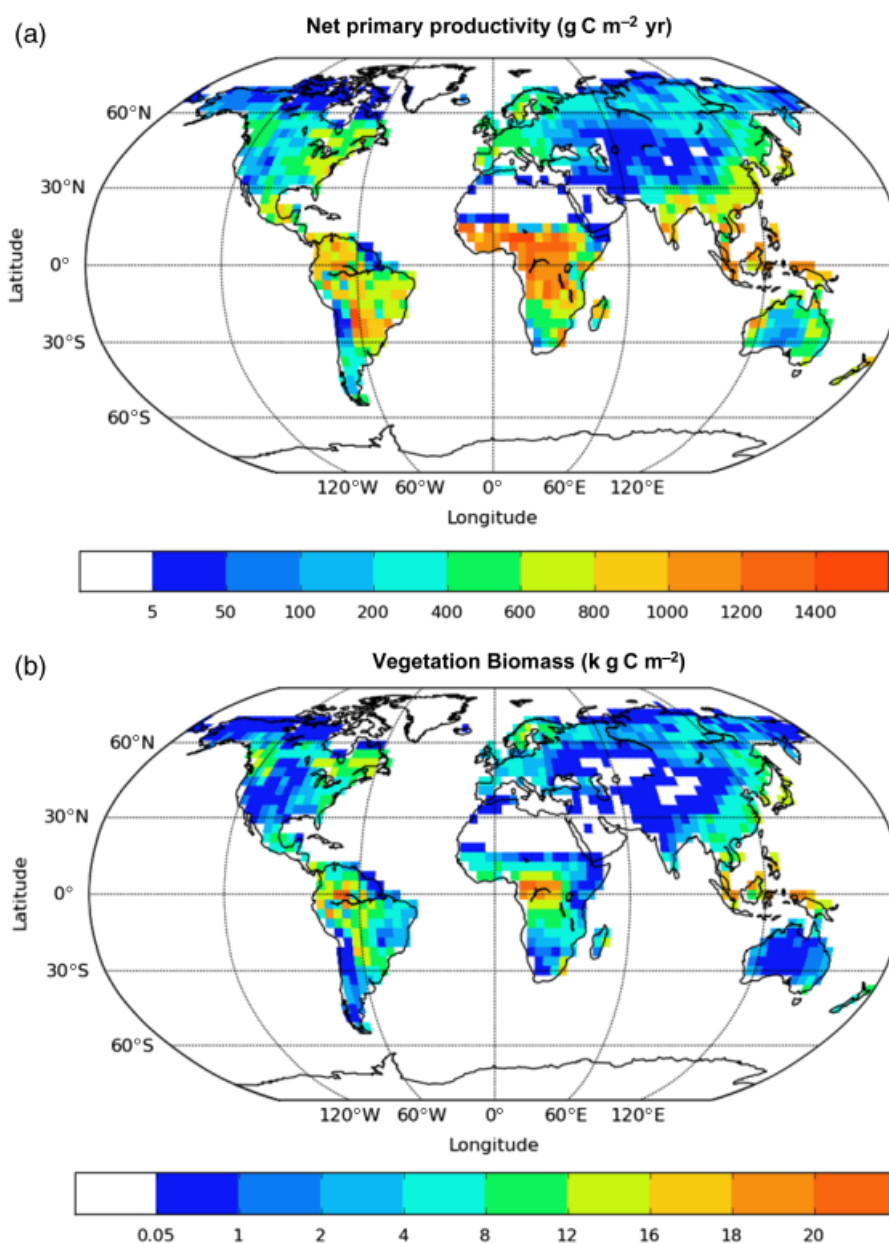


Fig. 5 Simulated net primary productivity (a) and vegetation biomass (b) for the preindustrial control run using RF99 based 1850 land cover.

vegetation biomass until about 1960 after which it begins to increase again because of the reduction in deforestation rates (Fig. 6a) and continuing CO₂ fertilization. Figure 7c indicates higher litter mass in simulations with LUC compared with those without because of the deforested biomass that enters the litter pool (see Table 1), although the changes are comparatively small. Figure 7d shows that simulations that include changes in crop area exhibit little change or decrease in soil carbon values because of the increased

soil carbon decomposition rates over cropland associated with tillage (red and orange lines). However, in simulations which also include the increase in pasture area (magenta and green lines) the soil carbon mass increases because of lower soil decomposition rates for grasslands. In terms of total terrestrial carbon storage (Fig. 7a), other than the pasture-only simulations, all simulations with LUC lie reasonably close to each other with increases in land carbon by year 2000 varying from 26 to 40 PgC which

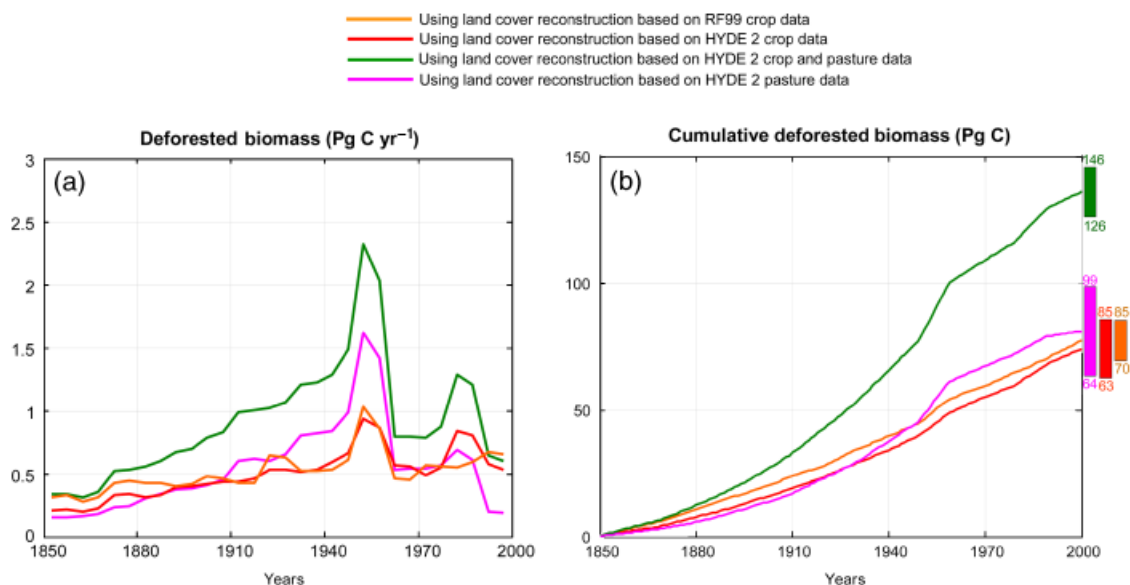


Fig. 6 Five-year moving average values of simulated deforested biomass (L) for the different land cover reconstructions (a) and their cumulative values $\int L dt$ (b). Values are averaged for the cases using reconstructions based on the linear and rule-based approaches. The bars in (b) show the range of cumulative emissions by year 2000 when using linear or the rule-based approaches. The lower values correspond to the rule-based approach.

is 50–65 PgC less than in simulations without LUC. In pasture-only simulations (magenta lines), the deforestation associated with the increase in pasture area is compensated, to a large extent, by an increase in soil carbon such that the total land carbon is only ~ 10 PgC less than the simulations without LUC. The effect of the inclusion of pasture together with crops is to yield about 13 PgC higher carbon accumulation in land (the difference between red and green lines in Fig. 7a). Overall, the use of HYDE and RF99 based crop data yields similar results (red and orange lines), while including pasture from the HYDE data yields slightly higher terrestrial carbon storage (green line) (Fig. 7a).

Atmospheric CO_2 . The differences in land cover reconstructions affect interactions between the earth system components which redistribute deforested carbon among the land, atmosphere and oceans. The resulting differences in atmospheric CO_2 are shown in Fig. 8 which compares the simulated globally averaged surface CO_2 with observation-based values shown in black. For clarity the results are grouped for simulations, with no LUC (blue), with LUC based on increases in pasture area only (magenta) and with LUC based on increases in crop and crop plus pasture area (orange). The shading indicates the range of values obtained for the simulations in these categories.

The simulations without LUC have less atmospheric CO_2 than simulations with LUC, as expected. The CO_2

concentration at year 2000 in these simulations is $\sim 353.7 \pm 2.7$ ppm which is 17.3 ± 6.3 ppm lower than in simulations with LUC. This difference can be compared with Matthews *et al.* (2004) (26 ppm) and Brovkin *et al.* (2004) (22–43 ppm) mentioned earlier. Simulations with increases in pasture area only yield CO_2 concentration of 356.6 ± 1.8 ppm at year 2000, whereas simulations with increases in crop area and both crop and pasture areas yield a CO_2 concentration of 371.1 ± 3.7 ppm at year 2000 and compare best with the observed value of 369.4 at Mauna Loa. Simulations with an increase in pasture area only yield lower atmospheric CO_2 concentration, despite similar deforested biomass, when compared with simulations with an increase in crop area only (Fig. 6a) because of large compensating carbon sequestration in soils (Fig. 7d).

CO_2 fluxes and LUC emissions. The changes in total land carbon in Fig. 7a are the cumulative result of atmosphere to land CO_2 fluxes (F_L) in Fig. 9 which also gives the Intergovernmental Panel on Climate Change (IPCC) Fourth Assessment Report (AR4) estimates of the flux and its uncertainty ranges for the 1980s and 1990s (Denman *et al.*, 2007). The neglect of LUC (blue line) results in a large simulated uptake of carbon during the 1980s and 1990s that lies outside the IPCC AR4 uncertainty range and which results in low simulated values of atmospheric CO_2 (Fig. 8). The use of

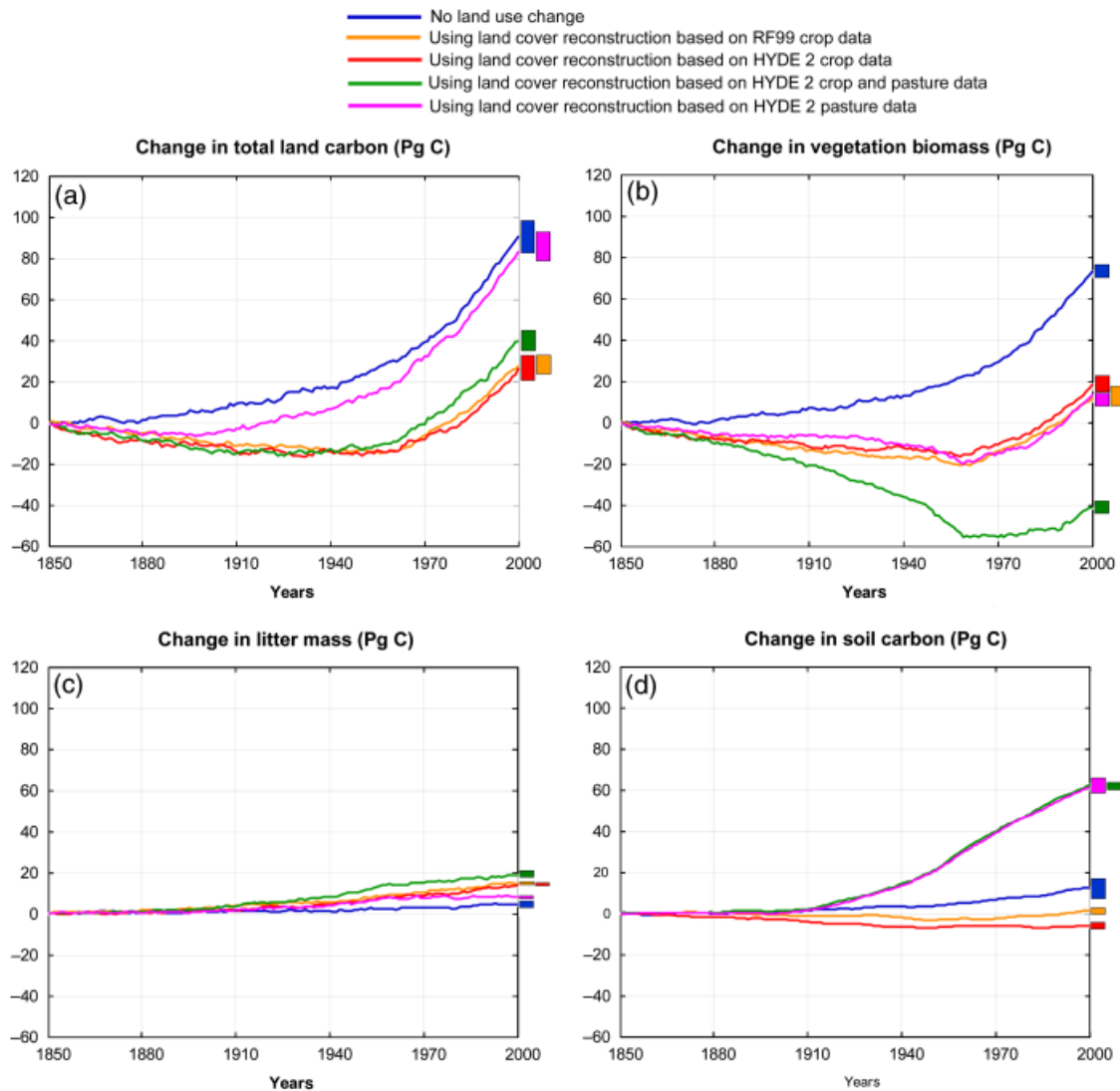


Fig. 7 Simulated 1850–2000 changes in total land carbon (a), vegetation biomass (b), litter mass (c) and soil carbon mass (d) for simulations with and without land use change (LUC). Values are averaged for the simulations with LUC for reconstructions based on the linear and rule-based approaches. The results for no-LUC simulations are averaged for the cases using RF99 and HYDE based 1850 land cover. The bars show the range at year 2000 when using the linear or the rule-based approach for simulations with LUC. The range for no-LUC simulations is due to the use of RF99 or HYDE based 1850 land cover.

land cover reconstructions with changes in pasture area only (magenta line) also results in large carbon uptake during the 1980s and 1990s. All other simulations yield reasonably similar results which lie within the IPCC AR4 uncertainty range, although the simulations with crop and pasture changes (green line) yield higher land carbon uptake than do crop-only simulations (red and orange lines).

In Fig. 9, the difference in atmosphere to land CO_2 flux between simulations with and without LUC yields the net LUC emissions for the different land cover reconstructions [Eqn (4)]. Figure 10a displays these LUC emissions for the 1850–2000 period together with

those from Houghton (2003). The LUC emissions from the model give the amount of carbon released from the land as a consequence of increases in cropland and/or pasture and replacement of natural vegetation as well as the response of the system to these LUCs. Figure 10b gives the cumulative LUC emissions to 2000 and the coloured bars show the range between results when using the linear and the rule-based approaches. The cumulative LUC emissions for crop-only and crop and pasture scenarios range from 40 to 76 Pg C, compared with 156 Pg C from Houghton (2003). The cumulative value for pasture-only scenarios are much lower because of carbon sequestration in soil. The implied

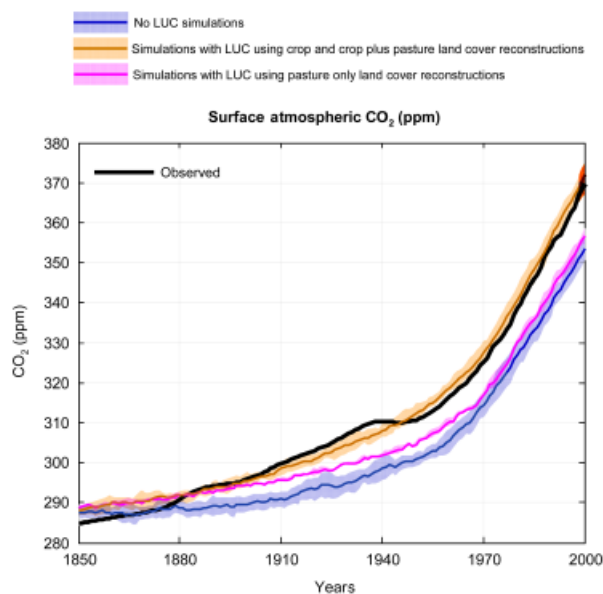


Fig. 8 Simulated surface atmospheric CO_2 concentration for the simulations with and without anthropogenic land use change (LUC) for the 1850–2000 period. Observed CO_2 is shown in black. The shaded area shows the range between similar simulations. The blue shaded region is from simulations A and D without LUC. The magenta shaded region is from simulations I and J with increase in pasture area only. The orange shaded regions is from simulations B, C, E, F, G and H with increase in crop area and both crop and pasture areas.

LUC emissions (E_L) indicate a source of $0.25\text{--}0.84 \text{ Pg C yr}^{-1}$ during the 1990s for simulations that are based on changes in crop-only and crop and pasture together, compared with the Houghton (2003) estimate of around 2.1 Pg C yr^{-1} (Fig. 10a). For pasture-only simulations, the net implied LUC emissions give a weak sink when using the linear approach ($\sim 0.02 \text{ Pg C yr}^{-1}$) and a weak source when using the rule-based approach ($\sim 0.29 \text{ Pg C yr}^{-1}$) during the 1990s (only averaged results are shown in Fig. 10a). Our 1990s estimates of LUC emissions are similar to those of Shevliakova *et al.* (2009) who find that the historical increase in croplands yields a source of $0.6\text{--}0.9 \text{ Pg C yr}^{-1}$ and that pasture increase yields fluxes that vary from a sink of 0.15 to a source of $0.37 \text{ Pg C yr}^{-1}$, although their cumulative totals are higher than ours.

Figure 10a also shows the range of LUC emissions in the McGuire *et al.* (2001) study for the 1920–1990 period from four TEMs that use RF99 data for deforestation, Houghton's book-keeping approach for the treatment of deforested biomass, and which are driven with specified observed CO_2 and climate. Our LUC estimates are lower than those of McGuire *et al.*

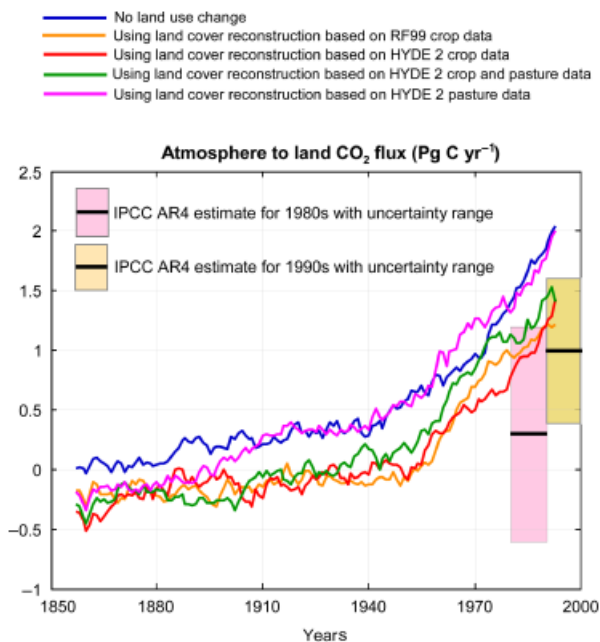


Fig. 9 Simulated 1850–2000 atmosphere to land CO_2 flux (F_L) for the simulations with and without land use change (LUC). Positive values indicate a flux of carbon into the land. IPCC AR4 values for the 1980s and 1990s are also shown with their uncertainty ranges. Values are averaged for the simulations with LUC for reconstructions based on the linear and rule-based approaches. The results for no-LUC simulations are averaged for the cases using RF99 and HYDE based 1850 land cover. Plotted values are fifteen year moving averages.

(2001), Jain & Yang (2005) and Shevliakova *et al.* (2009) primarily because in our approach CO_2 evolution is fully interactive and driven by emissions and LUC information. This aspect is discussed in more detail in Discussion.

Figure 11a shows the simulated atmosphere to ocean CO_2 flux (F_O) for the 1850–2000 period for the different land cover reconstructions together with 1980s and 1990s estimates and their uncertainty ranges from IPCC AR4. Figure 11b gives the cumulative values of F_O . Atmosphere to ocean CO_2 fluxes in these simulations respond primarily to atmospheric CO_2 . CanESM1 simulated ocean–atmosphere fluxes are somewhat lower than other estimates as discussed in Arora *et al.* (2009) because of slightly too warm ocean temperatures and somewhat weak vertical exchange possibly limited by model's coarse vertical resolution. For crop-only and crop and pasture land cover reconstructions, the cumulative 1850–2000 ocean uptake varies from 88 to 95 Pg C .

The cumulative increase in the atmospheric carbon burden (179.7 Pg C) over the 1850–2000 period is the result of the cumulative emissions (299.4 Pg C) partially

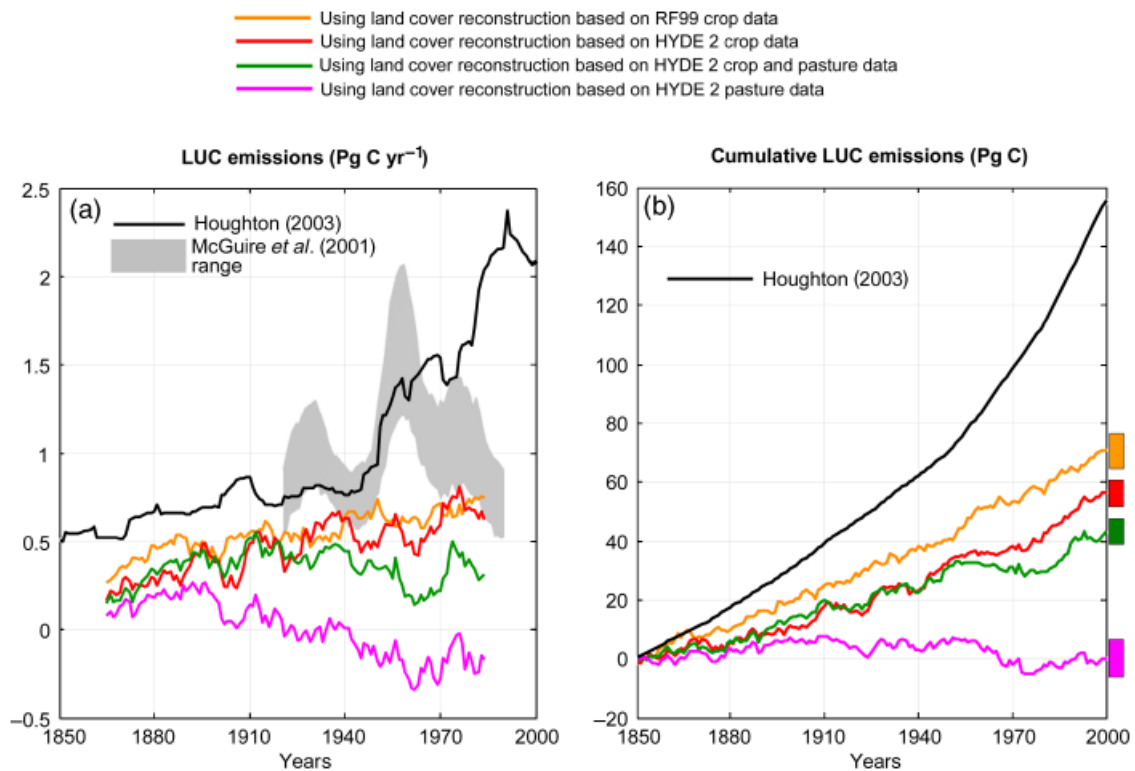


Fig. 10 Estimated land use change (LUC) emissions for the 1875–1985 period (a) and their cumulative values for the 1850–2000 period (b). Thirty-year moving averaged values are shown and results are averaged for the reconstructions based on the linear and rule-based approaches. The bars show the range at year 2000 when using the linear or the rule-based approach. Houghton (2003) emissions are also shown. Panel (a) additionally shows the range of LUC emissions simulated in the McGuire *et al.* (2001) study.

offset by land and ocean carbon uptake. Table 3 summarizes these quantities for simulations with and without LUC and compares them to observation-based estimates. The atmospheric carbon burden is known from the observed atmospheric CO_2 concentration. Observation-based estimates of cumulative carbon uptake by the global ocean are from Sabine *et al.* (2004) and Denman *et al.* (2007) and estimated fossil fuel and industrial CO_2 emissions are from Marland *et al.* (2007) as discussed in Arora *et al.* (2009). When calculated as a residual of these observation-based estimates, the cumulative uptake by land is -10 PgC which indicates that the land has been a net source of atmospheric CO_2 over the 1850–2000 period despite being a sink during the 1980s and 1990s (Denman *et al.*, 2007). Table 3 indicates that the modelled cumulative ocean uptake is lower than that of Sabine *et al.* (2004). However, Matsumoto & Gruber (2005) suggest that the Sabine *et al.* (2004) estimate may be high by $\sim 7\%$ or around 10 PgC which would make land nearly carbon neutral over the 1850–2000 period. The land carbon uptake in simulations using RF99 and HYDE 2 crop-only land cover reconstructions is the smallest among all simulations. The effect of an

increase in pasture area is to increase carbon uptake by land, decrease the atmospheric carbon burden and consequently decrease uptake by the ocean. The differences between results using linear and rule-based approaches are around 9 PgC for the cumulative land carbon uptake. Table 3 indicates that the observed atmospheric carbon burden change is broadly consistent with the LUC representation in the model except for the no-LUC and the pasture-only cases which yield lower than observed change in atmospheric carbon burden. This is also the case for the net land plus ocean uptake which is higher than observed for the no-LUC and pasture-only cases. Although the partitioning between land and ocean uptakes indicates some disagreement the net land plus ocean uptake for the crop and crop plus pasture cases agrees well with the observational-based values.

North–south structures

Figure 12 shows the zonally averaged atmosphere to land CO_2 flux (F_L) averaged over the 1980–2000 period for simulations with and without LUC. Results are shown for the no LUC case, for the case with increase

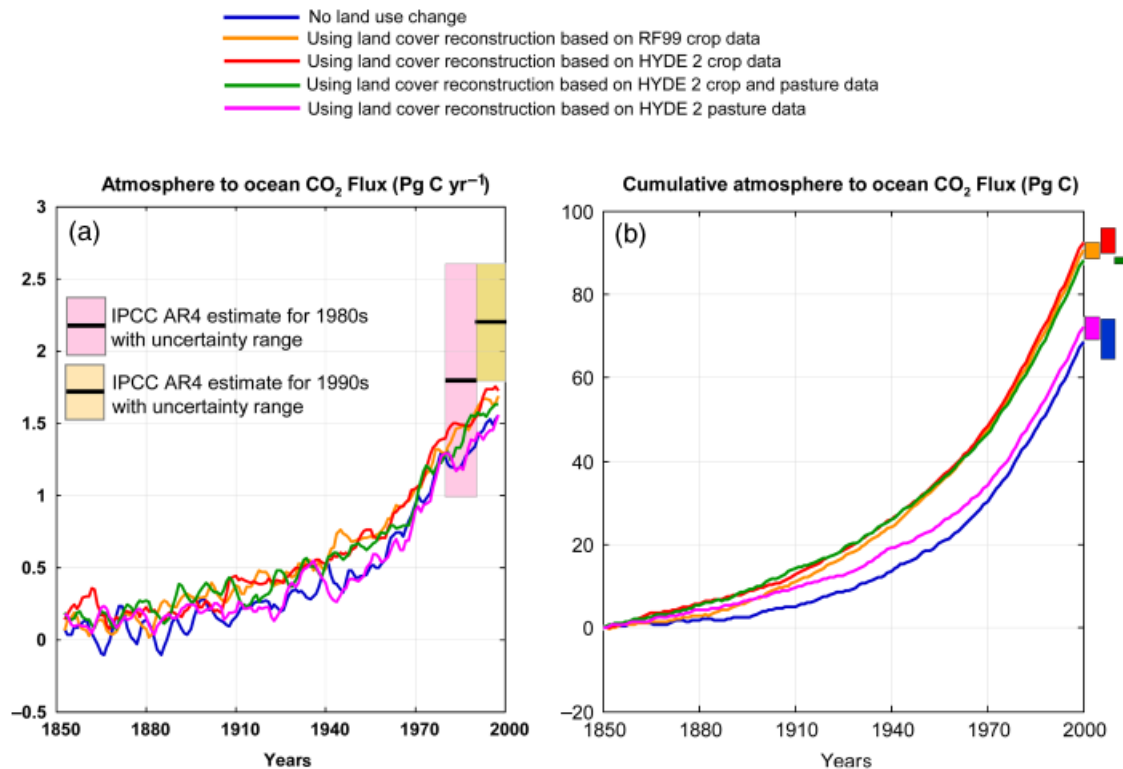


Fig. 11 Simulated 1850–2000 atmosphere to ocean CO₂ flux (F_O) for the simulations with and without land use change (LUC, a) and their cumulative values (b). Positive values indicate flux of carbon into the ocean. IPCC AR4 values for the 1980s and 1990s are also shown with their uncertainty ranges in (a). Values are averaged for the simulations with LUC for reconstructions based on the linear and rule-based approaches. The results for no-LUC simulations are averaged for the cases using RF99 and HYDE based 1850 land cover. Plotted values are 5-year moving averages. The coloured bars show the range at year 2000 when using the linear or the rule-based approach.

in crop area and for an increase in crop plus pasture area. In the absence of anthropogenic LUC (blue line), the land is a sink of atmospheric carbon at most latitudes except, of course, in regions where there is no vegetation. When LUC is characterized as an increase in crop area (red line) then F_L for tropical and southern hemisphere land becomes weaker and even locally negative while F_L for the mid- to high-latitude Northern hemisphere land remains positive and hence a carbon sink. These results are consistent with those reported in IPCC AR4 (Denman *et al.*, 2007) and other studies (Myneni *et al.*, 2001; Goodale *et al.*, 2002; Kaufmann & Stock, 2003; Stephens *et al.*, 2007). The effect of an increase in crop area is to reduce the capacity of land to take up atmospheric carbon almost everywhere (the red line is below the blue line). LUC characterized by change in pasture as well as crop area (green line) yields a sink of atmospheric carbon in the tropics and, as well, a large atmospheric carbon sink is simulated between 10°S and 30°S. This is not consistent with inversion-based studies that suggest the tropics are near neutral or a small carbon sink and is the result of the large pasture

increase in Australia in the HYDE data set as discussed below.

Geographical patterns

Figure 13 shows the spatial distribution of changes in total land carbon from 1850 to 2000 for a no-LUC simulation (using 1850 land cover based on RF99) and simulations that use different land cover reconstructions following the rule-based approach. In the absence of LUC, total land carbon increases almost everywhere (Fig. 13a), except in regions with little vegetation as seen also in the zonally averaged plots of Fig. 12. Land cover reconstructions based on an increase in crop area (Fig. 13b and c) lead to a decrease in land carbon in the Midwestern United States, parts of Europe, parts of tropical Asia and South Eastern China and increases generally elsewhere. Differences in detail are seen depending on the land cover reconstruction used. There is a greater carbon loss in South Eastern China and parts of tropical Asia in the RF99 based land cover reconstruction and greater carbon loss in Europe when

Table 3 The change in atmospheric CO₂ burden and carbon uptake by the land and the ocean for the 1850–2000 period for the simulations with and without LUC

	Simulation with ...					Observation-based estimate
	No LUC	RF99 crop	HYDE 2 crop-only	HYDE 2 crop and pasture	HYDE 2 pasture-only	
Change in atmospheric carbon burden	136.5–144.1	178.1–185.3	178.5–183.5	167.3–175.1	140.6–148.3	179.7
Land uptake	83.3–98.2	23.8–31.9	21.5–31.2	35.1–44.4	76.0–90.3	–10.3 ± ?
Ocean uptake	64.6–72.8	89.5–91.8	90.0–95.1	87.7–88.7	70.2–74.3	130 ± 19 115 ± 17 (excluding extrapolation polewards of 65°)
Land and ocean uptake	156.1–162.7	115.6–121.3	116.6–121.1	123.8–132.1	150.3–160.5	119.7 ± ?

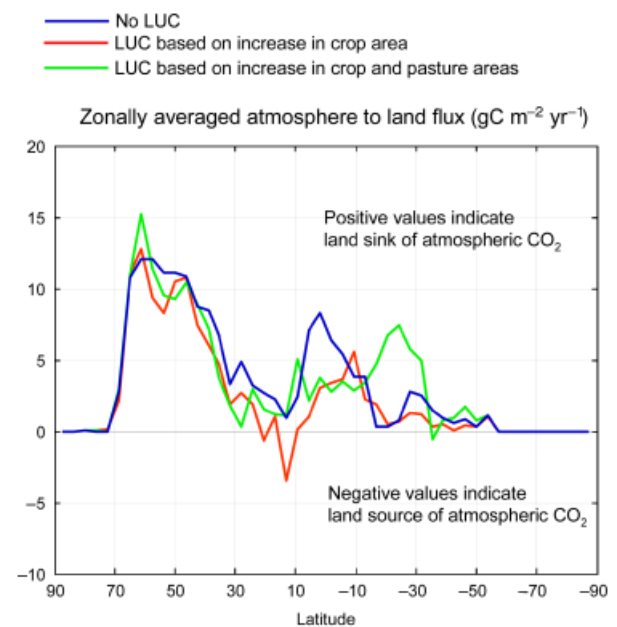
Fossil fuel, cement manufacturing and biofuel emissions are 299.4 Pg C in all cases. Observation-based values are also shown. The range is due to use of the linear or rule-based approach for land cover reconstruction for simulations with LUC and due to use of RF99 or HYDE 2 based 1850 land cover for simulations without LUC. Units are Pg C.

LUC, land use change.

HYDE 2 crop data is used. Both crop simulations yield a sink of carbon in the Eastern United States (Fig. 13b and c) due to abandonment of croplands. Using a land cover reconstruction that includes only an increase in pasture area results in land carbon increases over Eastern Siberia, the middle eastern region and most notably Australia (Fig. 13d) (seen also at 25°S in Fig. 12) because of what is likely an unrealistic increase in pasture area (Fig. 3h and k). Parts of South America, Western United States and South Eastern China see a loss of land carbon due to an increase in pasture area. Figure 13e shows changes in land carbon for increases in both in crop and pasture area based on the HYDE 2 data set and reflect the broad scale features of changes due to an increase in cropland area (Fig. 13c) and in pasture area (Fig. 13d). The spatial distribution of simulated changes in land carbon show how an increase in crop area over the 1850–2000 period has led to reductions in carbon storage since croplands store less carbon than the natural vegetation they replace. An increase in pasture area, however, is simulated to yield both increases (due to higher soil carbon storage) and decreases (due to deforestation) in land carbon storage in different areas of the world.

Discussion

There are at least three ways of estimating the CO₂ emissions associated with changes in land use. The book-keeping approach of Houghton (2003) assigns deforested biomass to various pools which decay at specified rates that are insensitive to changes in CO₂ and climate. A somewhat more interactive approach uses a TEM in an offline mode driven with specified

**Fig. 12** Simulated zonal distribution of atmosphere to land CO₂ flux (F_L) averaged for the 1980–2000 period.

CO₂ and climate parameters for simulations with and without LUC (McGuire *et al.*, 2001; Jain & Yang, 2005; Shevliakova *et al.*, 2009). A fully interactive approach, as employed in this study, uses a comprehensive fully coupled carbon–climate model where the climate and carbon budget interact in simulations both with and without LUC. Differences in estimated LUC emissions between these approaches are due to differences in the manner deforestation is calculated, either directly as in Houghton (2003) or inferred indirectly on the basis of

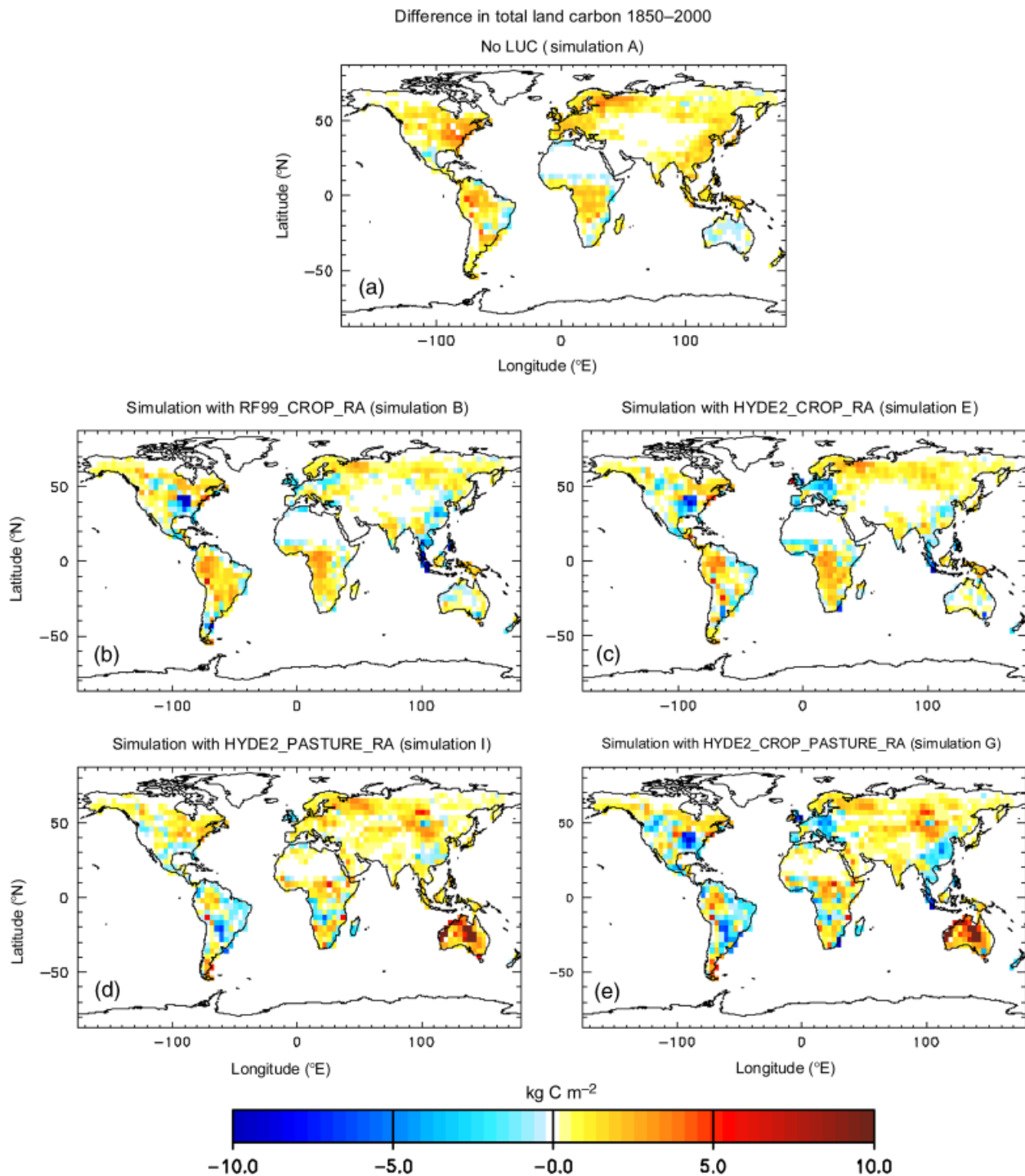


Fig. 13 Spatial distributions of simulated changes in total land carbon from 1850 to 2000 for a simulation without land use change (LUC) (using 1850 land cover based on RF99) and simulations with different land cover reconstructions obtained using the rule-based approach.

changes in crop and/or pasture area; differences in the vegetation biomass that is deforested; whether or not the effect of CO₂ increase and climate change is taken into account; and whether simulations with and with-

out LUC are driven with and see the same CO₂ and/or climate.

The direct Houghton (2003) approach for estimating LUC emissions does not account for climate and carbon

feedbacks due to changes in vegetation, albedo and atmospheric CO₂ concentration. The use of TEMs accounts for some of these feedbacks and the result is to reduce the inferred LUC emissions (McGuire *et al.*, 2001; Jain & Yang, 2005; Shevliakova *et al.*, 2009). The results obtained here using a fully interactive approach, where all feedbacks are acting, reduce inferred LUC emissions even further.

The difference in inferred LUC emissions between the fully interactive and TEM approaches can arise in several ways. If the same CO₂ concentration is specified for both the LUC and no-LUC simulations, for instance, the result will be to overestimate the LUC emissions. This is because the undisturbed vegetation in the no-LUC case is able to take up atmospheric CO₂ more efficiently because of the CO₂ fertilization effect. In terms of Eqn (4), $G > G^*$ and this increase in GPP acts to enhance the inferred LUC emissions. This is the primary reason why LUC emissions calculated here are lower than those of McGuire *et al.* (2001), Jain & Yang (2005) and Shevliakova *et al.* (2009) who force their offline simulations with the same atmospheric CO₂ concentrations for cases with and without LUC.

This effect is illustrated in Fig. 14. The fully interactive LUC emissions of Fig. 10, based on RF99 land cover reconstruction (thin orange line), are compared with inferred LUC emissions for a case where the no-LUC simulation is forced with the same atmospheric CO₂ as in the fully interactive LUC case. The result is the thick orange line in Fig. 14a giving larger inferred LUC emissions. In Fig. 14b, the cumulative LUC emissions are enhanced in this case by about 50 PgC to a value of 124 PgC compared with 71 PgC. This increase is due to the specification of the same CO₂ concentration for the simulation without LUC as was obtained in the simulation with LUC. The calculated LUC emissions (thick orange line) now also lie within the range of the McGuire *et al.* (2001) estimates.

Figure 14 also gives inferred LUC emissions calculated from simulations with CTEM (purple line) when driven with specified observed CO₂ concentrations and 1979–1999 reanalysis climate data (used repeatedly) for the 1850–1992 period using the rule-based RF99 land cover reconstruction. This methodology is same as McGuire *et al.* (2001), where simulations with and without LUC are driven with the same CO₂ as well as the same climate. The results in Fig. 14a show the same temporal evolution (e.g. emissions peaking during the mid-1950s) and magnitude as the range of the McGuire *et al.* (2001) results.

In regards to the historical increase in pasture area, Houghton (2003) attributes about 15% of global LUC emissions to changes in pasture area. Our results suggests that inclusion of pasture together with crops in the

HYDE 2 based data set does not increase the overall implied LUC emissions but rather reduces them because soil carbon sequestration in pasture lands dominates carbon loss from deforestation. The simulated carbon sequestration in pasture lands is a function of lower soil carbon decomposition rate for grasslands but also the manner in which land cover is reconstructed. Treating all pastures as grasslands, as is done here, represents an upper limit to the pasture-related LUC effects. Soil carbon decomposition rates for grassland ($\sim 0.013 \text{ year}^{-1}$) in CTEM are about 50% lower than that for other noncrop PFTs ($\sim 0.020\text{--}0.026 \text{ year}^{-1}$). A higher soil carbon decomposition rate for grasslands will also lower sequestration in soil carbon. Overall, even if carbon sequestration in grasslands is reduced by 25% for the 1850–2000 period (from $\sim 60 \text{ PgC}$, for green and magenta lines, to $\sim 45 \text{ PgC}$ in Fig. 7d) the total change in land carbon (Fig. 7a), and hence the net LUC emissions, will remain similar for simulation with increase in pasture and cropland area (green line) and simulation with increase in cropland area only (red and orange lines). Finally, the role of pastures in our results differs from that of Strassmann *et al.* (2008), who also use HYDE pasture data, and who estimate that the inclusion of pastures nearly doubles the estimated LUC emissions. Strassmann *et al.* (2008) do not mention the carbon sequestration potential of pasture/grasslands that leads to a carbon sink in our study and that of Shevliakova *et al.* (2009).

Summary and conclusions

LUC emissions are uncertain and Ramankutty *et al.* (2007) put a $\pm 50\%$ error band on values estimated by Houghton (2003). Here we investigate the uncertainty in LUC emissions and the associated uncertainty in the atmospheric and land carbon budgets using a comprehensive fully coupled climate and carbon cycle model. Eight reconstructions of 1850–2000 land cover are used, based on RF99 and HYDE 2 data sets of crop and pasture area changes. The LUC emissions are obtained by differencing net atmosphere to land CO₂ fluxes between simulations with and without anthropogenic LUC. The simulated cumulative deforested biomass varies from 63 to 145 PgC for the 1850–2000 period and cumulative LUC emissions vary from 40 to 77 PgC for the land cover reconstructions based on an increase in crop area and crop plus pasture areas. These cumulative LUC emissions (40–77 PgC) are lower than Houghton (2003) (156 PgC) as well as the new estimate of 148 PgC (Houghton, 2008). The resulting atmospheric CO₂ concentration in year 2000 varies between 367.3 and 374.7 ppm compared with the value of 369.4 ppm at Mauna Loa. Overall, the atmospheric

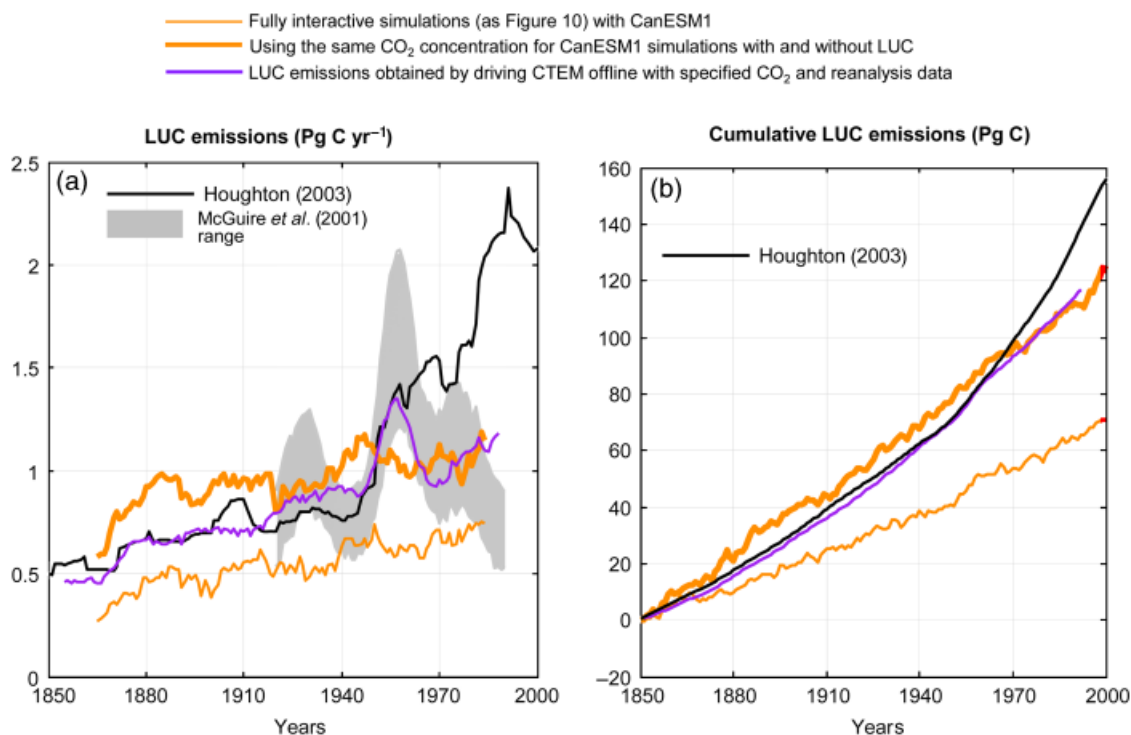


Fig. 14 Comparison of (a) calculated land use change (LUC) emissions and (b) their cumulative values for the RF99 based land cover reconstructions when with and without LUC simulations see different (thin orange line) and same (thick orange line) atmospheric CO₂ concentrations as explained in the text. Also shown are the estimated LUC emissions from offline simulations where CTEM is driven with specified observed CO₂ and repeated 1979–1999 reanalysis climate data repeatedly (purple line). Panel (a) also shows the range of LUC emissions simulated in the McGuire *et al.* (2001) study.

CO₂ in simulations with LUC changes is 17.3 ± 6.3 ppm higher than in simulations without LUC, depending on the data set used and the manner in which historical land cover is constructed.

Land cover reconstructions based on an increase only in pasture area yield cumulative LUC emissions varying between -7 and 7 PgC, i.e. a small sink or a small source, depending on whether the land cover is reconstructed using the linear or the rule-based approach. The overall emissions for pasture-only scenarios are lower because LUC emissions created from deforestation are compensated by an increase in soil carbon in pasture lands. The effect of pasture when included with crops is to decrease overall LUC emissions for the 1850–2000 period by ~ 10 to 15 PgC, again due to carbon sequestration in soils. The effect of increasing pasture area will depend on the location of the pastures and the manner in which land cover is reconstructed. The large increase in pasture area in the HYDE 2 data set when interpreted as increase in grassland area, especially over Australia, seems unrealistic and leads to a carbon sink between 10°S and 30°S that is not consistent with inversion-based studies (Denman *et al.*, 2007). The results also show that LUC emissions are overestimated

when simulations with and without LUC are performed with the same specified CO₂ concentrations independent of the treatment of pasture area.

Although fossil fuel emissions are relatively well constrained the same is not true for LUC emissions as they depend on interactive climate processes and atmospheric CO₂ concentration. Chapter 7 of IPCC AR4 (Denman *et al.*, 2007) notes that LUC emissions from McGuire *et al.* (2001) are lower than those of Houghton (2003) after 1970 (see Fig. 10a), but concludes that it is premature to say that Houghton's numbers are overestimated. Our analysis indicates that LUC emissions are reduced even further than those of McGuire *et al.* (2001) when calculated with in the framework of a coupled climate carbon cycle model. Estimated lower LUC emissions also implicitly imply that benefits of reduced emissions from prevented deforestation and degradation (REDD) (Miles & Kapos, 2008) may be lower than previously assumed.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Appendix S1. Rule-based and linear approaches for land cover reconstruction.

Figure S1. Schematic representation of rule-based and linear approaches for incorporating increase in crop area (panels a and b) and abandonment of croplands (panel c). Arrows with numbers 1 and 2 in panels (a) and (b) indicate the order in which demand for increased crop is satisfied. Arrows in panel (c) indicate how cropland abandonment leads to expansion of grasses which, if present, are later supplemented or replaced by woody vegetation.

Table S1. The order in which the demand for increased cropland is satisfied by decreasing the fractional coverage of noncrop PFTs, and the order in which abandoned croplands revert back to natural vegetation in the rule-based approach of Wang *et al.* (2006).

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